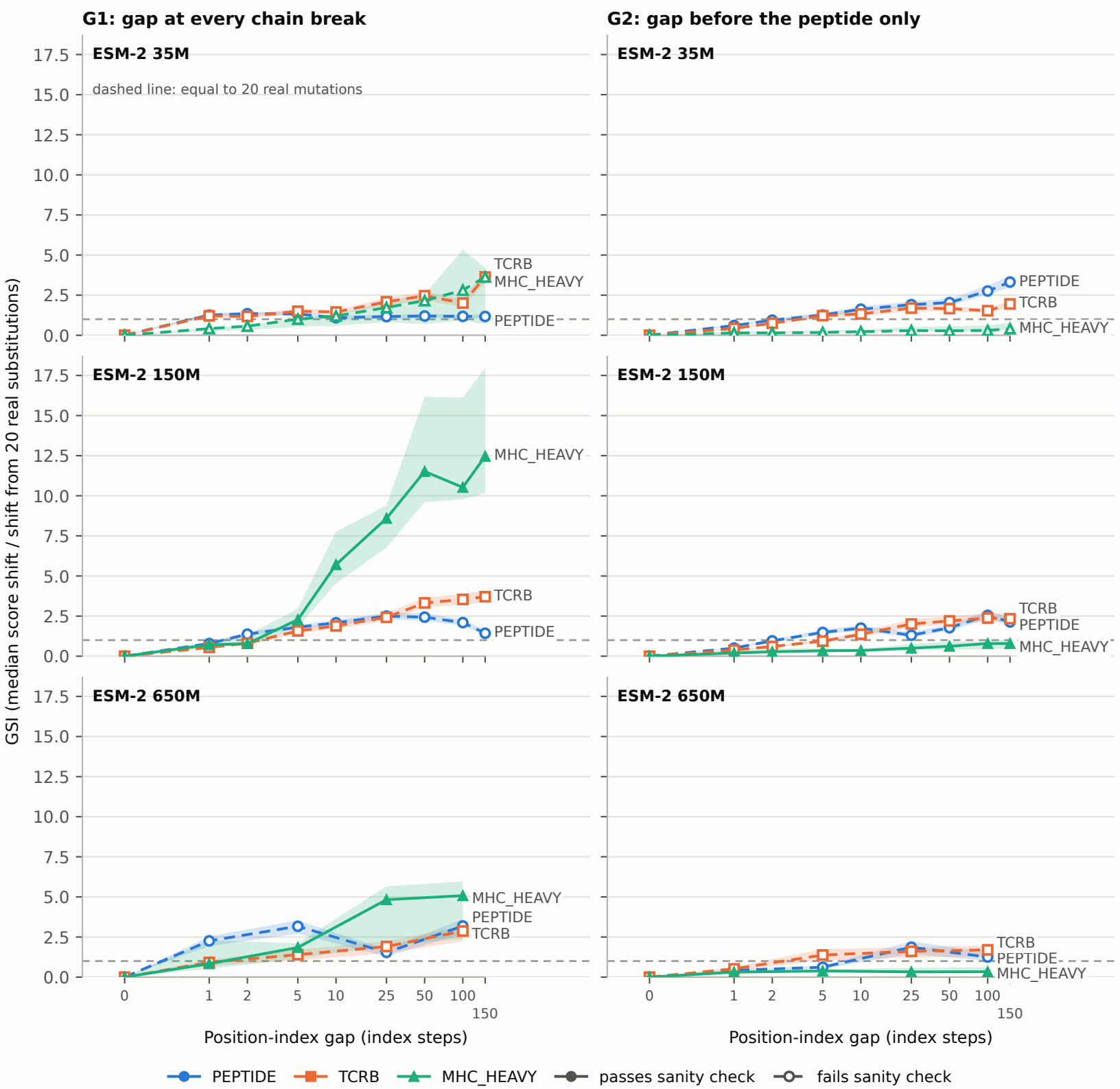
A content-free change to position indices moves ESM-2's score several times more than 20 real substitutions

Shown for MHC at 150M and 650M, the conditions whose estimator passes its sanity check; peptide and TCR spans are shown but not admissible.



MHC_HEAVY intervals: cluster bootstrap over alleles, Kish n_eff = 1.92 at 35M and 150M, 1.96 at 650M. PEPTIDE and TCRB intervals: example-level bootstrap. Shaded bands are 95% intervals.